

APPROXIMATIONS FOR THE NON-CENTRAL WISHART DISTRIBUTION

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1. Introduction

The non-central Wishart distribution was first derived by Anderson and Girshick (1944) and Anderson (1946) for the linear and planar cases. Herz (1955) expressed the distribution as a Bessel function and James (1961) as a series of zonal polynomials. This non-central distribution plays an important part in deriving a number of test-statistics in multivariate analysis. The distribution as well as the distributions derived from it, are difficult to handle.

In this paper it is shown how the non-central distribution can be approximated by a central distribution or by a central distribution together with the generalised Laguerre polynomials. The central distribution as used here, is the distribution of $A = \sum_{\alpha=1}^n X_{\alpha} X_{\alpha}'$, where the independent vectors X_{α} all have the same p -dimensional normal distribution $N(0, \Sigma)$. The probability density function is given by (Anderson (1958)):

$$(1.1) \quad \frac{|\Lambda|^{\frac{1}{2}(n-p-1)} \exp \{-\frac{1}{2} \text{tr} \Sigma^{-1} A\}}{2^{\frac{1}{2}np} \pi^{\frac{1}{2}p(p-1)} |\Sigma|^{\frac{1}{2}n} \prod_{i=1}^p \Gamma[\frac{1}{2}(n+1-i)]}$$

The non-central distribution is the distribution of $B = \sum_{\alpha=1}^n X_{\alpha} X_{\alpha}'$, where the independent vectors X_{α} each have a p -dimensional normal distribution $N(\mu_{\alpha}, \Sigma)$ and where the vectors μ_{α} are not all zero. The exact distribution of B is not needed in this paper. Only the first and second order moments and the differential equation in the moment generating function are needed. A straightforward method is therefore used to derive a differential equation in the moment function from first principles. This method is similar to that previously used by Steyn (1951) to derive the moment generating function of the central distribution.

2. A differential equation in the moment function of the non-central distribution

To simplify the notation the matrix D with elements d_{ij} will be used for Σ^{-1} .

The elements of X_α , that is $x_{1\alpha}, x_{2\alpha}, \dots, x_{p\alpha}$, then has the joint probability function:

$$f_\alpha \equiv f_\alpha(x_{1\alpha}, x_{2\alpha}, \dots, x_{p\alpha}) = C \exp \left[-\frac{1}{2} \sum_{i,j=1}^p d_{ij} (x_{i\alpha} - \mu_{i\alpha})(x_{j\alpha} - \mu_{j\alpha}) \right]$$

with $d_{ij} = d_{ji}$, C a constant and $\mu_{1\alpha}, \mu_{2\alpha}, \dots, \mu_{p\alpha}$ the elements of the vector μ_α .

The moment generating function of the joint distribution of the elements b_{rs} of B is by definition:

$$(2.1) \quad M \equiv M(\theta_{11}, \theta_{12}, \dots, \theta_{pp}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left[\prod_{\alpha=1}^n f_\alpha \right] \exp \left[\sum_{r \leq s} b_{rs} \theta_{rs} \right] \prod_{i=1}^p \prod_{\alpha=1}^n dx_{i\alpha}$$

Differentiate (2.1) w.r.t. θ_{ij} , then

$$(2.2) \quad \frac{\partial}{\partial \theta_{ij}} M = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left[\prod_{\alpha=1}^n f_\alpha \right] \exp \left[\sum_{r \leq s} b_{rs} \theta_{rs} \right] b_{ij} \prod_{i=1}^p \prod_{\alpha=1}^n dx_{i\alpha}$$

$$\text{with } b_{ij} = \sum_{\alpha=1}^n x_{i\alpha} x_{j\alpha}.$$

Putting

$$\sum_{j=1}^p d_{ij} x_{j\alpha} = \sum_{j=1}^p d_{ij} (x_{j\alpha} - \mu_{j\alpha}) + \sum_{j=1}^p d_{ij} \mu_{j\alpha}$$

it follows from (2.2) that

$$\begin{aligned} \sum_{j=1}^p d_{ij} \frac{\partial}{\partial \theta_{ij}} M &= \sum_{\alpha=1}^n \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{\beta \neq \alpha} f_\beta \left\{ \int_{-\infty}^{\infty} \exp \left[\sum_{r \leq s} b_{rs} \theta_{rs} \right] \right. \\ &\quad \left. \cdot \left(\sum_{j=1}^p d_{ij} (x_{j\alpha} - \mu_{j\alpha}) f_\alpha \right) x_{i\alpha} dx_{i\alpha} \right\} \prod_{j \neq i\alpha} \prod_{\beta \neq i\alpha} dx_{j\beta} \\ &+ \sum_{\alpha=1}^n \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left[\prod_{\alpha=1}^n f_\alpha \right] \exp \left[\sum_{r \leq s} b_{rs} \theta_{rs} \right] \sum_{j=1}^p d_{ij} \mu_{j\alpha} x_{i\alpha} \prod_{i=1}^p \prod_{\alpha=1}^n dx_{i\alpha}. \end{aligned}$$

Using $\sum_{j=1}^p d_{ij} (x_{j\alpha} - \mu_{j\alpha}) f_\alpha = -\frac{\partial}{\partial x_{i\alpha}} f_\alpha$ it follows on integrating by parts w.r.t. $x_{i\alpha}$ the function inside the curly brackets and remembering that f_α vanishes at its boundaries that,

$$\sum_{j=1}^p d_{ij} \frac{\partial}{\partial \theta_{1j}} M = \delta_{i1} n M + \sum_{j=1}^p \Delta_{ij} \theta_{ij} \frac{\partial}{\partial \theta_{1j}} M + \sum_{j=1}^p d_{ij} \sum_{\alpha=1}^n \mu_{j\alpha} X_{1\alpha}$$

where $\delta_{i1} = 1 (i=1), \Delta_{ij} = 2 (i=j)$

$$= 0 (i \neq 1) = 1 (i \neq j)$$

and

$$X_{1\alpha} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \left[\prod_{\alpha=1}^n f_{\alpha} \right] \exp \left[\sum_{r \leq s} b_{rs} \theta_{rs} \right] x_{1\alpha} \prod_{j=1}^p \prod_{\alpha=1}^n dx_{j\alpha}.$$

Now $p-1$ further sets of differential equations follow similarly by differentiation w.r.t. $\theta_{2j}, \dots, \theta_{pj}$. The general set can be written as

$$(2.3) \quad \sum_{j=1}^p d_{ij} \frac{\partial}{\partial \theta_{\ell j}} M = \delta_{i\ell} n M + \sum_{j=1}^p \Delta_{ij} \theta_{ij} \frac{\partial}{\partial \theta_{\ell j}} M + \sum_{j=1}^p d_{ij} \sum_{\alpha=1}^n \mu_{j\alpha} X_{\ell\alpha}$$

where $\delta_{i\ell} = 1 (i = \ell)$ for $i, \ell = 1, 2, \dots, p$.
 $= 0 (i \neq \ell).$

Furthermore, again putting

$$\sum_{j=1}^p d_{ij} x_{j\alpha} \mu_{\ell\alpha} = \sum_{j=1}^p d_{ij} (x_{j\alpha} - \mu_{j\alpha}) \mu_{\ell\alpha} + \sum_{j=1}^p d_{ij} \mu_{j\alpha} \mu_{\ell\alpha}$$

so that

$$\sum_{j=1}^p d_{ij} \sum_{\alpha=1}^n X_{j\alpha} \mu_{\ell\alpha} = \sum_{\alpha=1}^n \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{\beta \neq \alpha} f_{\beta} \left\{ \int_{-\infty}^{\infty} \exp \left[\sum_{r \leq s} b_{rs} \theta_{rs} \right] \cdot \left(\sum_{j=1}^p d_{ij} (x_{j\alpha} - \mu_{j\alpha}) f_{\alpha} \right) \mu_{\ell\alpha} dx_{j\alpha} \right\} \prod_{j \neq i} \pi dx_{j\beta} + M \sum_{j=1}^p d_{ij} \sum_{\alpha=1}^n \mu_{j\alpha} \mu_{\ell\alpha}.$$

it follows as before, after integrating the function inside the curly brackets, by parts w.r.t. $x_{i\alpha}$ that

$$(2.4) \quad \sum_{j=1}^p d_{ij} \sum_{\alpha=1}^n X_{j\alpha} \mu_{\ell\alpha} = \sum_{j=1}^p \Delta_{ij} \theta_{ij} \sum_{\alpha=1}^n X_{j\alpha} \mu_{\ell\alpha} + M \sum_{j=1}^p d_{ij} \sum_{\alpha=1}^n \mu_{j\alpha} \mu_{\ell\alpha}$$

for $i, \ell = 1, 2, \dots, p$.

Using the notation,

$$\mu_{\alpha} = \begin{pmatrix} \mu_{1\alpha} \\ \mu_{2\alpha} \\ \vdots \\ \mu_{p\alpha} \end{pmatrix}, \quad X_{\alpha} = \begin{pmatrix} X_{1\alpha} \\ X_{2\alpha} \\ \vdots \\ X_{p\alpha} \end{pmatrix}$$

and

$$D = \begin{pmatrix} d_{11} & d_{12} & \dots & d_{1p} \\ d_{12} & d_{22} & \dots & d_{2p} \\ \vdots & & & \vdots \\ d_{1p} & d_{2p} & \dots & d_{pp} \end{pmatrix}, \quad \theta = \begin{pmatrix} \theta_{11} & \theta_{12} & \dots & \theta_{1p} \\ \theta_{12} & \theta_{22} & \dots & \theta_{2p} \\ \vdots & & & \vdots \\ \theta_{1p} & \theta_{2p} & \dots & \theta_{pp} \end{pmatrix}$$

and

$$\frac{\partial}{\partial \theta} M = \begin{pmatrix} \frac{\partial}{\partial \theta_{11}} & \frac{\partial}{\partial \theta_{12}} & \dots & \frac{\partial}{\partial \theta_{1p}} \\ \frac{\partial}{\partial \theta_{12}} & \frac{\partial}{\partial \theta_{22}} & \dots & \frac{\partial}{\partial \theta_{2p}} \\ \vdots & & & \vdots \\ \frac{\partial}{\partial \theta_{1p}} & \frac{\partial}{\partial \theta_{2p}} & \dots & \frac{\partial}{\partial \theta_{pp}} \end{pmatrix} M,$$

it follows in matrix notation from (2.3) and (2.4) that

$$(2.5) \quad (D - \theta) \frac{\partial}{\partial \theta} M = n M I + \sum_{\alpha=1}^n D \mu_{\alpha} X'_{\alpha}$$

and

$$(D - \theta) \sum_{\alpha=1}^n X_{\alpha} \mu'_{\alpha} = M D \sum_{\alpha=1}^n \mu_{\alpha} \mu'_{\alpha}$$

where I denotes the unit matrix. Substituting for $\sum_{\alpha=1}^n \mu_{\alpha} X'_{\alpha}$ or $(\sum_{\alpha=1}^n X_{\alpha} \mu'_{\alpha})'$ in (2.5) it

follows that

$$(2.6) \quad (D-\theta) \frac{\partial}{\partial \theta} M = (n I + D T D (D-\theta)^{-1}) M$$

$$\text{where } T = \sum_{\alpha=1}^n \mu_{\alpha} \mu'_{\alpha}$$

For the central case (1.1) T is the null matrix and the equation (2.6) becomes

$$(2.7) \quad (D-\theta) \frac{\partial}{\partial \theta} M = n I M.$$

3. Some properties derived from the differential equation

Since $(D-\theta)$ is non-singular (2.6) can be written as

$$(3.1) \quad \frac{\partial}{\partial \theta} M = (n I (D-\theta)^{-1} + (D-\theta)^{-1} D T D (D-\theta)^{-1}) M.$$

Thus, the expected value of b_{rs} is

$$\begin{aligned} E[b_{rs}] &= \left[\frac{\partial}{\partial \theta_{rs}} M \right] \theta_{ij} = 0 \text{ for all } i \text{ and } j \text{ values.} \\ &= n \frac{|D_{rs}|}{|D|} + t_{rs} \end{aligned}$$

$$(3.2) \quad = n \sigma_{rs} + t_{rs}$$

where $|D_{rs}|$ is the cofactor of d_{rs} in $|D|$, σ_{rs} is an element of Σ and

$$T = \begin{pmatrix} t_{11} & t_{12} & \cdots & t_{1p} \\ t_{12} & t_{22} & \cdots & t_{2p} \\ \vdots & & & \vdots \\ t_{1p} & t_{2p} & \cdots & t_{pp} \end{pmatrix}$$

is symmetrical.

The second order moments follow from (3.1) as

$$\begin{aligned} E[b_{rs} b_{vw}] &= \left[\frac{\partial}{\partial \theta_{vw}} \frac{\partial}{\partial \theta_{rs}} M \right] \theta_{ij} = 0 \text{ for all } i, j. \\ (3.3) \quad &= (n \sigma_{vw} + t_{vw})(n \sigma_{rs} + t_{rs}) + n[\sigma_{wr} \sigma_{vs} + \sigma_{vr} \sigma_{ws}] + \sigma_{wr} t_{vs} \\ &\quad + \sigma_{vr} t_{ws} + \sigma_{vs} t_{wr} + \sigma_{ws} t_{vr} \end{aligned}$$

The covariance between b_{rs} and b_{vw} is given by:

$$\begin{aligned} E[b_{rs} - (n\sigma_{rs} + t_{rs})][b_{vw} - (n\sigma_{vw} + t_{vw})] \\ = n(\sigma_{wr}\sigma_{vs} + \sigma_{vr}\sigma_{ws}) + \sigma_{wr}t_{vs} + \sigma_{vr}t_{ws} + \sigma_{vs}t_{wr} + \sigma_{ws}t_{vr}. \end{aligned}$$

For $r = v$ and $s = w$ it follows that:

$$\text{Var.}(b_{rs}) = n(\sigma_{rs}^2 + \sigma_{rr}\sigma_{ss}) + 2\sigma_{rs}t_{rs} + \sigma_{rr}t_{ss} + \sigma_{ss}t_{rr}.$$

4. Approximation of the non-central distribution by a central distribution

The moments of the central distribution (1.1) can be written down from the corresponding moments for the non-central distribution by taking T to be a null matrix i.e. $t_{rs} = 0$ for all r and s values. From (3.2) and (3.3), it therefore follows for the elements of A that

$$E(a_{rs}) = n\sigma_{rs}$$

and

$$E(a_{rs}a_{vw}) = n^2\sigma_{vw}\sigma_{rs} + n(\sigma_{wr}\sigma_{vs} + \sigma_{vr}\sigma_{ws}).$$

Instead of (1.1) a new distribution is now considered, namely the distribution of A where in the parent normal population the covariance matrix is changed to $\Sigma^* = \Sigma + \frac{1}{n}T$ and where the different $\mu_{i\alpha}$'s are small compared with the variances and where T is defined as in (2.6). Let σ_{rs}^* denote an element of this covariance matrix, then $\sigma_{rs}^* = \sigma_{rs} + \frac{1}{n}t_{rs}$. The first order moments of this new central distribution are then given by:

$$\begin{aligned} E[a_{rs}] &= n\sigma_{rs}^* \\ (4.1) \quad &= n\sigma_{rs} + t_{rs}, \end{aligned}$$

which are the same as the first order moments of the non-central distribution as derived in (3.2).

The second order moments of this new distribution are given by:

$$E[a_{rs}a_{vw}] = n^2\sigma_{vw}^*\sigma_{rs}^* + n(\sigma_{wr}^*\sigma_{vs}^* + \sigma_{vr}^*\sigma_{ws}^*)$$

$$(4.2) \quad = (n\sigma_{vw} + t_{vw})(n\sigma_{rs} + t_{rs}) + n(\sigma_{wr}\sigma_{vs} + \sigma_{vr}\sigma_{ws}) \\ + \sigma_{wr}t_{vs} + \sigma_{vr}t_{ws} + \sigma_{vs}t_{wr} + \sigma_{ws}t_{vr} + \frac{1}{n}(t_{wr}t_{vs} + t_{vr}t_{ws}).$$

It now follows from (3.3) that the second order moments of the new distribution differ from those of the non-central distribution in the last term of order $O(n^{-1})$ only.

The results (4.1) and (4.2) suggest that the probability density function for the non-central distribution of B may be approximated by a function similar to (1.1) but with Σ replaced by $\Sigma^* = \Sigma + \frac{1}{n}T$, that is, by the function

$$(4.3) \quad f(B) = \frac{|B|^{\frac{1}{2}(n-p-1)} \exp[-\frac{1}{2}\text{tr}(\Sigma + \frac{1}{n}T)^{-1}B]}{2^{\frac{1}{2}np} \pi^{\frac{1}{2}p(p-1)} |\Sigma + \frac{1}{n}T|^{\frac{1}{2}n} \prod_{i=1}^p \Gamma[\frac{1}{2}(n+1-i)]}$$

To show that this is true the non-central differential equation (3.1) which can be written as

$$(4.4) \quad \frac{\partial}{\partial \theta} M = n[(1-\Sigma\theta)^{-1}\Sigma + (1-\Sigma\theta)^{-1}(\frac{1}{n}T)\Sigma^{-1}(1-\Sigma\theta)^{-1}\Sigma]M$$

will now be compared with the differential equation in the moment function for the central distribution (4.3) which, by (2.7), is given as

$$(4.5) \quad \frac{\partial}{\partial \theta} M = n[(\Sigma + \frac{1}{n}T)^{-1} - \theta]^{-1}M.$$

Clearly, by taking θ such that conditions for convergence of matrix series are satisfied, it follows from (4.5) that

$$\begin{aligned} \frac{\partial}{\partial \theta} M &= n \left[1 - (\Sigma\theta + \frac{1}{n}T\theta) \right]^{-1} (\Sigma + \frac{1}{n}T) M \\ &= n \left[1 + (\Sigma\theta + \frac{1}{n}T\theta) + (\Sigma\theta + \frac{1}{n}T\theta)^2 + (\Sigma\theta + \frac{1}{n}T\theta)^3 + \dots \right] \\ &\quad \cdot (\Sigma + \frac{1}{n}T) M \\ &= n \left[(1-\Sigma\theta)^{-1} + (1-\Sigma\theta)^{-1} \frac{1}{n}T\theta(1-\Sigma\theta)^{-1} + O(n^{-2}) \right] \\ &\quad \cdot (\Sigma + \frac{1}{n}T) M \end{aligned}$$

where $O(n^{-2})$ represent matrices with elements containing a factor n^{-k} where $k \geq 2$.

Thus

$$(4.6) \quad \frac{\partial}{\partial \theta} M = n [(1-\Sigma\theta)^{-1} \Sigma + (1-\Sigma\theta)^{-1} \left(\frac{1}{n}T\right) \Sigma^{-1} (1-\Sigma\theta)^{-1} \Sigma + O(n^{-2})] M.$$

The expressions in the brackets of equations (4.4) and (4.6) differ only in terms of $O(n^{-2})$, which indicates the closeness of the approximation of the non-central distribution of B by (4.3). It follows immediately from (4.4) and (4.6) that the moment generating function in the non-central case can differ from the moment generating function of (4.3) only in terms of $O(n^{-1})$.

It is logical to consider better approximations for the non-central distribution by determining further terms in the above calculations leading to (4.6) and to obtain series solutions in terms of the moment generating function of the central distribution (4.3). However, the algebra involved in this procedure is complicated and therefore a second approximation is considered in terms of the generalised Laguerre polynomials.

5. Approximation for the non-central distribution by using generalised Laguerre polynomials

Starting with the function $f(B)$ as defined in (4.3) a series expansion for the non-central probability density function $g(B)$ can be written as (Herz (1955), p. 502)

$$(5.1) \quad g(B) = \{1 + \alpha L_2^{\frac{1}{2}(n-p-1)}(B^*) + \beta L_1^{\frac{1}{2}(n-p-1)}(B^*) + O(n^{-3})\} f(B)$$

where the L -functions are generalised Laguerre polynomials (Constantine (1966)) defined in terms of zonal polynomials and are orthogonal with respect to $f(B)$ as weight function where

$$(5.2) \quad L_2^{\frac{1}{2}(n-p-1)}(B^*) = \left(\frac{1}{2}n\right)_2 C_2(l_p) \sum_{j=0}^2 \sum_J a_{2,J} \frac{C_J(-B^*)}{\left(\frac{1}{2}n\right)_J C_J(l_p)},$$

$$(5.3) \quad L_1^{\frac{1}{2}(n-p-1)}(B^*) = \left(\frac{1}{2}n\right)_1 C_1(l_p) \sum_{j=0}^2 \sum_J a_{1,J} \frac{C_J(-B^*)}{\left(\frac{1}{2}n\right)_J C_J(l_p)}$$

and

$$B^* = \frac{1}{2}(\Sigma + \frac{1}{n}T)^{-1}B$$

Using the orthogonal properties of these Laguerre polynomials it is immediately clear from (5.1) that the first order moments of $g(B)$ and of $f(B)$ are the same. To determine α and β in terms of the moments of $g(B)$, multiply (5.1) first by $L_2^{\frac{1}{2}(n-p-1)}(B^*)$

(B^*) and then by $L_2^{\frac{1}{2}(n-p-1)}(B^*)$ and integrate so that,

$$\int_{B>0} L_2^{\frac{1}{2}(n-p-1)}(B^*) g(B) d B = \alpha \int_{B>0} [L_2^{\frac{1}{2}(n-p-1)}(B^*)]^2 f(B) d B$$

giving

$$\alpha = \frac{E[L_2^{\frac{1}{2}(n-p-1)}(B^*)]}{\int_{B>0} [L_2^{\frac{1}{2}(n-p-1)}(B^*)]^2 f(B) d B}$$

and similarly

$$\beta = \frac{E[L_1^{\frac{1}{2}(n-p-1)}(B^*)]}{\int_{B>0} [L_1^{\frac{1}{2}(n-p-1)}(B^*)]^2 f(B) d B}$$

The coefficients α and β are functions of the first and second order moments of the non-central variable B . We leave further approximation for a later communication.

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